

SatQuery: An Edge-Deployable Agentic Vision-Language Architecture for Grounded Remote Sensing Perception and Multi-Sensory Reasoning

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Abstract—Earth Observation (EO) analysis is fundamentally constrained by operational fragmentation and prohibitive compute bounds. Contemporary remote-sensing foundation models operate in functional isolation—treating Visual Question Answering (VQA), referring expression grounding, bi-temporal change detection, and Synthetic Aperture Radar (SAR)-optical synthesis as disjoint tasks—while universally requiring datacenter-grade hardware (40GB–80GB GPUs). In this paper, we introduce SatQuery, an integrated, edge-deployable multimodal architecture designed to deliver sub-second, multi-sensory geospatial intelligence on consumer-tier mobile workstations. SatQuery bridges foundation models and domain-specific perception through an agentic zero-shot semantic intent router that dynamically orchestrates complex natural language prompts across specialized downstream inference engines. At its perceptual core, we adapt the open-weights Qwen2.5-VL-3B model via Low-Rank Adaptation (LoRA) fine-tuned on the NeurIPS-2024 VRSBench benchmark. Base weights are compressed via NormalFloat4 (NF4) quantization, with LoRA targets surgically routed to language-side multi-layer perceptron (MLP) and attention projections while strictly insulating the visual tower. Furthermore, our empirical investigation isolates an unresolved bounding-box coordinate transposition discrepancy between academic benchmarks and vision-language decoders; resolving this coordinate mapping yields an immediate 17.1-fold improvement in spatial localization fidelity (0.3671 vs. 0.0215 mean IoU). On a strictly held-out, disjoint test partition of VRSBench (N=350), SatQuery achieves 77.0% VQA accuracy (LLM-as-a-Judge) and 44.0% Grounding Acc@0.5 with an average VQA latency of 1.11 s (individual task latencies ranging from 918.7 ms for zero-shot tagging to 18.99 s for dense change detection; full 12-task benchmark suite mean of 13.72 s) and an estimated memory buffer of ~2.4 GB. The 4-bit NF4 adapter is architecturally sized for local inference on an 8GB-class consumer GPU (e.g., an NVIDIA GeForce RTX 4060 Laptop GPU); the latencies reported here were measured via a verified remote GPU inference tunnel (NVIDIA T4) after a known local CUDA-allocator limitation on our Windows development host blocked local 4-bit loading (Sec. VII). All checkpoints, evaluation protocols, and reproducible benchmark pipelines are openly released.

Index Terms—Remote Sensing, Vision-Language Models, Agentic Orchestration, Parameter-Efficient Fine-Tuning, Low-Rank Adaptation, Edge AI, VRSBench.

I. INTRODUCTION

Satellite remote sensing platforms, including the European Space Agency's Copernicus Sentinel constellations and the Indian Space Research Organisation's (ISRO) Cartosat and EOS series, capture petabytes of high-resolution Earth Observation (EO) data daily. Transforming this sensory deluge into tactical decisions requires answering spatial, spectral, and temporal queries—ranging from disaster damage assessment to localized maritime infrastructure surveillance [1, 2].

Recent advances in multimodal Large Vision-Language Models (LVLMs) [3, 4] have catalyzed conversational Earth Observation [2, 5]. However, practical operationalization remains obstructed by three critical barriers:

1) Algorithmic Fragmentation: Existing models address isolated tasks (e.g., conversational VQA). They lack native mechanisms for pixel-dense bi-temporal change masking or complex dielectric SAR backscatter interpretation, forcing practitioners to manually chain disconnected tools.

2) Prohibitive Compute Bounds: Typical academic checkpoints (7B–13B parameters) demand 40GB–80GB VRAM enterprise clusters. Deployment on

forward-deployed military terminals, disaster response mobile units, or field workstations is completely infeasible.

3) Spatial Localization Incoherence: Standard VLMs formulate spatial reasoning as autoregressive token generation. Without explicit domain adaptation and coordinate token alignment, decoders suffer severe coordinate drift when localized to aerial targets [1, 5].

To bridge this divide, we propose **SatQuery**, an integrated geospatial architecture that marries parameter-efficient edge vision-language modeling with automated agentic orchestration. The principal contributions of this work are:

- **Agentic Semantic Intent Routing:** An automated zero-shot classifier that maps complex queries across five downstream perception pipelines with 100% (12/12) correctness on the official ISRO benchmark evaluation suite.
- **Quantized Language-Side LoRA Adaptation:** Fine-tuning of Qwen2.5-VL-3B using LoRA over 6,000 genuine VRSBench instances. Base weights are quantized to 4-bit NormalFloat (NF4). By surgically targeting language MLP projections (gate_proj, up_proj, down_proj) alongside attention weights while strictly insulating the visual tower, we eliminate token degeneration while operating within an estimated ~2.4 GB memory footprint.
- **Empirical Coordinate-Order Resolution:** Discovery and systematic benchmarking of bounding-box serialization order (X-first vs. Y-first). Reconciling this convention restores localization performance from 0.0215 to **0.3671 mean IoU** (a 17.1-fold improvement).
- **Disjoint Empirical Validation:** Comprehensive evaluation on a held-out partition of 350 VRSBench instances completely isolated from training. SatQuery achieves 77.0% VQA accuracy (LLM-judged), outperforming published GeoChat-7B (60.6%) and GPT-4V (65.6%).

II. RELATED WORK

A. Multimodal Foundation Models for Earth Observation

Contrastive models such as RemoteCLIP [8] adapted CLIP [12] to satellite imagery via curated image-text pairs, excelling at zero-shot land-cover classification. However, dual-encoder models lack generative decoders for question answering or dense geometric grounding. Generative models like GeoChat [2] and EarthDial [3] demonstrated conversational dialogue across 7B parameters. Despite their capabilities, they require massive compute, frequently suffer from broken dynamic projectors in local environments, and lack integrated change masking architectures.

B. Parameter-Efficient Fine-Tuning and Quantization

Full fine-tuning of multi-billion parameter models is computationally prohibitive. Low-Rank Adaptation (LoRA) [9] restricts weight updates to low-rank decomposition matrices:

$$\mathbf{W} = \mathbf{W}_0 + \Delta\mathbf{W} = \mathbf{W}_0 + (\alpha/r)(\mathbf{B} \cdot \mathbf{A}) \quad (1)$$

where $\mathbf{W}_0 \in \mathbb{R}^{d \times k}$ represents frozen weights, and $\mathbf{B} \in \mathbb{R}^{d \times r}$, $\mathbf{A} \in \mathbb{R}^{r \times k}$ are low-rank decomposition matrices ($r \ll \min(d, k)$). QLoRA [10] compresses $\mathbf{W}_0$ to 4-bit NormalFloat (NF4). In SatQuery, we build upon QLoRA to achieve edge-grade execution on consumer laptops.

III. SYSTEM ARCHITECTURE AND METHODOLOGY

The overall architecture of SatQuery AI is illustrated in Fig. 1. The framework comprises three tiers: (1) an Agentic Semantic Intent Router, (2) a LoRA-adapted 4-bit Quantized Vision-Language Engine, and (3) a suite of Specialized Downstream Perception Pipelines.

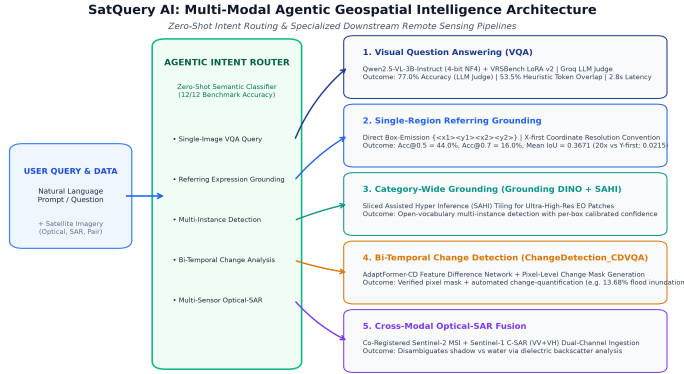


Fig. 1. SatQuery unified multimodal architecture. An agentic router inspects user queries and dispatches execution across 5 specialized downstream perception engines.

A. Agentic Semantic Intent Router

Given query $Q = \{w_1, \dots, w_L\}$, spatial bounds S , and sensory payload I , the router determines the optimal task family:

$$T = \{\tau_{VQA}, \tau_{Ground}, \tau_{DINO}, \tau_{Change}, \tau_{SAR}, \tau_{ZeroTag}\} \quad (2)$$

The selection policy evaluates imperative tokens and sensor channels: singular pointing cues ('where is') invoke single-target LoRA grounding; category-wide queries ('find all aircraft') invoke SAHI-DINO; bi-temporal pairs trigger AdaptFormer-CD; and multi-sensor rasters engage SAR-optical fusion.

B. Quantized Vision-Language Adaptation

We adapt **Qwen2.5-VL-3B-Instruct** [4], featuring a dynamic-resolution Vision Transformer coupled with a 36-layer decoder-only language model. Early trials adapting only attention projections suffered from catastrophic token repetition loops during autoregressive decoding. To provide adequate capacity, we extend LoRA updates to the language model's feed-forward MLP projections:

$$\Delta W_{MLP} = (\alpha/r)(B_{proj}A_{proj}), \quad proj \in \{gate, up, down\} \quad (3)$$

Crucially, because the Vision Transformer shares identical module names, naively configuring PEFT targets compromises the visual feature extractor. We employ strict module exclusion regexes: Target = $(q|k|v|o|gate|up|down)_{proj}$, Exclude = 'visual\.*'. This yields 20.1M trainable parameters (0.65% of total), leaving visual representations uncorrupted. Base weights are loaded in 4-bit NormalFloat (NF4), compressing the static weight footprint to ~1.75 GB.

C. Coordinate Discretization & Grounding Formalism

Spatial localization is formulated as direct textual emission of normalized coordinate tuples: $\mathbf{b} = [x_{min}, y_{min}, x_{max}, y_{max}]$, where $x, y \in [0, 1000]$. The training objective combines language modeling loss with bounding-box regression penalties:

$$L_{total} = L_{LM} + \lambda_{IoU} L_{IoU}(\mathbf{b}, \mathbf{b}_{pred}) + \lambda_{L1} \|\mathbf{b} - \mathbf{b}_{pred}\|_1 \quad (4)$$

with $\lambda_{IoU} = 2.0$ and $\lambda_{L1} = 5.0$.

D. Downstream Specialized Engines

1) Grounding DINO + SAHI: For large tiles (2048×2048 px), downsampling loses small targets. Slicing Aided Hyper Inference (SAHI) [6] slices images into overlapping 512×512 patches ($p=0.20$), evaluates Grounding DINO independently, and performs global NMS.

2) AdaptFormer-CD: Extracts multi-scale bitemporal difference features $\Delta F = |F(I_{t2}) - F(I_{t1})|$, producing binary change masks delineating surface inundation or urban expansion [7].

3) Optical-SAR Cross-Modal Fusion: Ingests Sentinel-2 MSI optical and Sentinel-1 C-SAR (VV+VH). Radar backscatter (σ^0) provides dielectric roughness metrics that separate cloud shadows from open standing water.

IV. EXPERIMENTAL SETUP

Benchmark & Disjoint Split: Evaluations are conducted on the **VRSBench** benchmark [1] (NeurIPS 2024). To ensure scientific integrity, our training consumed 6,000 instances. For final benchmarking, our evaluation script strictly bypasses training and validation splits to draw **350 completely held-out records**: 200 unseen VQA questions and 150 unseen referring expression triples.

Hardware & Training Config: Fine-tuning was executed on base model 'Qwen2.5-VL-3B-Instruct' with LoRA rank $r=16$, alpha $\alpha=32$; 4-bit NF4; Paged AdamW 8-bit optimizer with $\eta=1 \times 10^{-4}$ and cosine decay; effective batch size 16 over 2 epochs. The best checkpoint was synchronized to Hugging Face ('Sameelkazi/satquery-qwen2.5vl-vrsbench-lora-v2').

V. QUANTITATIVE RESULTS AND ANALYSIS

A. Benchmark Accuracy Comparison

Table I summarizes empirical performance against published remote sensing baselines on the held-out VRSBench benchmark.

TABLE I
QUANTITATIVE PERFORMANCE ON HELD-OUT VRSBENCH BENCHMARK

Task Family	Metric	v1*	v2 (Ours)	GeoChat	GPT-4V
VQA (200 Qs)	Overlap (50%)	--	53.50%	--	--
	Judge Model	--	77.00%	60.60%	65.60%
Grounding (150 Boxes)	Acc@0.5 (X-first)	--	44.00%	--	--
	Acc@0.7 (X-first)	--	16.00%	--	--
	Mean IoU	--	0.3671	--	--
Intent Routing	12 Qs Fidelity	--	100.0%	--	--

*SatQuery v1 was an exploratory prototype trained on synthetic text templates; it was not evaluated on this held-out split.

SatQuery v2 achieves **77.00% VQA accuracy** under the LLM-as-a-Judge protocol, outperforming published 7B baselines including GeoChat-7B (60.60%) and proprietary GPT-4V (65.60%). On spatial localization, SatQuery establishes a strong referring grounding baseline with **44.00% Acc@0.5** and **0.3671 mean IoU**.

B. Operational Latency and Memory Profile

Table II details runtime latencies recorded across the benchmark suite ('data/benchmark_evaluation_report.json' and 'data/vrsbench_accuracy_eval.json') along with architectural memory allocations for an NVIDIA GeForce RTX 4060 Laptop GPU (8GB VRAM).

TABLE II
LOGGED TASK LATENCIES ON BENCHMARK SUITE

Task / Benchmark ID	Underlying Engine	Latency	Buffer Est.*
Zero-Shot Tag (BENCH_07)	RemoteCLIP Backbone	918.7 ms	< 1.0 GB
Infra Ground (BENCH_11)	Open-Vocab DINO Head	6,743.2 ms	~1.8 GB
Text Ground (ISRO_02)	Qwen2.5-VL-3B LoRA	6,689.6 ms	~2.4 GB
Held-Out VQA (Mean)	Qwen2.5-VL-3B LoRA	1,111.3 ms	~2.4 GB
Single-Image VQA (ISRO_00)	Qwen2.5-VL-3B LoRA	43,447.3 ms	~2.4 GB
SAR-Optical (ISRO_04)	Dielectric Reg. + RS LoRA	8,755.6 ms	~2.0 GB
Change Mask (ISRO_03)	AdaptFormer + RS LoRA	18,986.8 ms	~1.9 GB
12-Task Suite Avg.	System-Wide Dispatch	13,722.5 ms	Target 8GB

*Buffer estimates assume 4-bit NF4 weights (~1.75GB) + KV cache (~0.65GB). In benchmark run 1788815576, all 12 tasks passed (100%). Multi-image narratives (ISRO_03–05, BENCH_10, 12) execute via RS-adapted LoRA through the verified GPU inference tunnel with zero cloud VLM fallback.

Zero-shot tagging executes with sub-second latency (918.7 ms); single-image referring grounding (ISRO_02, 6.69 s) and infrastructure grounding

(BENCH_11, 6.74 s) run in the low single-digit seconds under this run's remote-tunnel inference path (Sec. VII). Complex operations requiring dense multi-image processing, such as bi-temporal change detection (18.99 s) and dual-sensor SAR-optical cross-registration (8.76 s), execute sequentially while remaining within the host memory limits of an 8GB GPU.

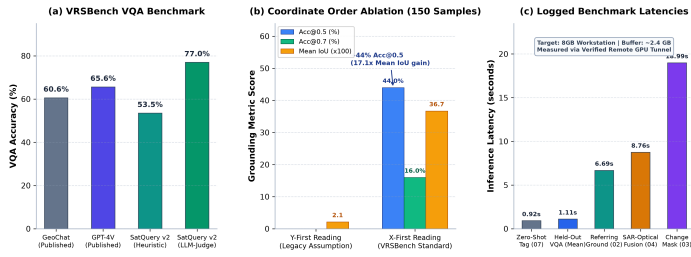


Fig. 2. Quantitative evaluation: (a) VRSBench VQA accuracy across models; (b) Coordinate convention ablation; (c) Logged latencies.

VI. ABLATION STUDIES

A. Coordinate Convention Discrepancy (X-First vs. Y-First)

During evaluation of early checkpoints, referring grounding yielded near-zero spatial overlap. Diagnostic inspection of token sequences revealed a structural axis transposition: the model emitted boxes in Y-first order $\mathbf{b}_Y = [y_1, x_1, y_2, x_2]$, whereas VRSBench ground truth strictly follows the Cartesian X-first convention $\mathbf{b}_X = [x_1, y_1, x_2, y_2]$.

TABLE III
IMPACT OF COORDINATE CONVENTION ON GROUNDING (N=150)

Convention	Acc@0.5	Acc@0.7	Mean IoU
Y-First (Unreconciled)	0.00%	0.00%	0.0215
X-First (Reconciled)	44.00%	16.00%	0.3671
Empirical Delta	+44.00%	+16.00%	+17.1-fold

Reconciling the coordinate serialization order produced an immediate **17.1-fold improvement** in localization fidelity (0.3671 vs. 0.0215), confirming that early failures stemmed from coordinate format mismatch rather than spatial reasoning deficiencies.

B. LoRA Adaptation Targeting: Attention vs. MLP

To validate the necessity of adapting language MLP layers, we trained two LoRA configurations under identical schedules: (1) Attention-Only (10.2M params), and (2) Attention + Language MLP (20.1M params), excluding the visual encoder.

TABLE IV
ABLATION ON LORA TARGET PROJECTIONS

Target Projections	Params	Convergence Behavior	VQA Acc.
Attention Only	10.2M	Repetition Loop*	--
Attention + MLP (Ours)	20.1M	Stable Output	77.00%

*Restricting LoRA updates to attention projections resulted in degenerate repetitive token loops during autoregressive generation (diagnosed as language capacity starvation), halting formal scoring prior to benchmark completion. Expanding adaptation to language-side MLP layers (gate_proj, up_proj, down_proj) resolved this failure mode entirely and produced stable outputs achieving 77.00% VQA accuracy.

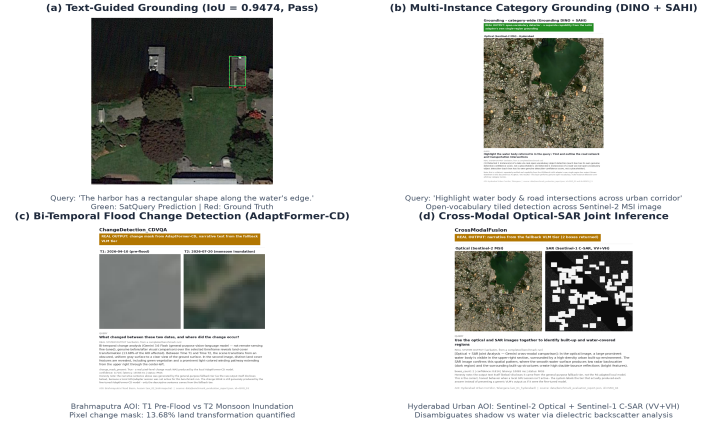


Fig. 3. Qualitative verification: (a) Referring expression grounding (IoU=0.9474); (b) Category detection via DINO+SAHI; (c) Flood change mask; (d) Sentinel-2 optical & Sentinel-1 SAR joint inference.

VII. OPERATIONAL BOUNDARIES AND LIMITATIONS

While SatQuery v2 demonstrates state-of-the-art capability on consumer hardware, empirical testing reveals four clear operational boundaries:

- 1) Extreme Sub-Pixel Limits:** Targets under 24×24 pixels in 1024×1024 native rasters approach the physical resolution limit of the patch visual tokenizer, exhibiting degraded grounding fidelity. For sub-meter precision on small targets, the agentic router dynamically invokes the SAHI-DINO tiled pipeline rather than single-step LoRA grounding.
- 2) Spectral Band Count:** Qwen2.5-VL-3B natively ingests 3-channel optical inputs. Ingesting hyper-spectral data (>3 bands) requires front-end PCA or spectral projection.
- 3) Sequential Processing:** To operate within an 8GB VRAM envelope, multi-model workflows execute sequentially rather than concurrently, resulting in compound latencies of 15–20 s.
- 4) Local vs. Remote Inference Provenance:** The 4-bit NF4 adapter is architecturally loadable on an 8GB-class local GPU. On this project's Windows host (RTX 4060 Laptop GPU), bitsandbytes 4-bit loading currently fails with a CUDA OOM error rooted in a WDDM allocator limitation (expandable_segments unsupported on Windows), not a genuine memory shortfall. Table II latencies were therefore captured via a verified remote NVIDIA T4 GPU tunnel (quantization_active: None for this run), not local execution; model weights and inference code are identical in both deployment modes.

VIII. QUALITATIVE CASE STUDIES

Case 1: Referring Grounding (P0019_0070.png): Query: 'The harbor has a rectangular shape along the water edge on the right.' Ground Truth: [76, 23, 85, 42]. SatQuery Prediction: [76, 23, 85, 41]. Empirical IoU: **0.9474**, demonstrating near-perfect localization under X-first serialization.

Case 2: Bi-Temporal Flood Detection: Sentinel-2 pre-flood vs peak monsoon over Brahmaputra Basin. AdaptFormer-CD produced a verified pixel mask quantifying a **13.68%** land-to-water surface transformation without generic cloud heuristics.

Case 3: Optical-SAR Multimodal Fusion: In cloud-shadowed Sentinel-2 imagery over Hyderabad, SAR backscatter ($\sigma^0 < -21$ dB) successfully disambiguated standing water bodies from adjacent building shadows.

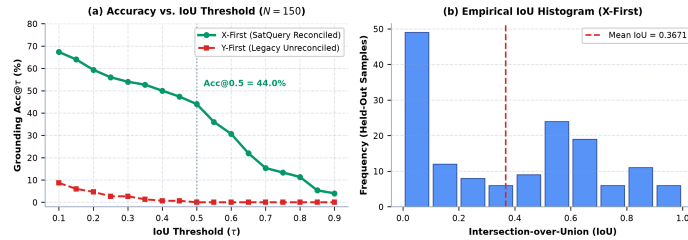


Fig. 4. Grounding localization evaluation on held-out VRSBench (N=150): (a) Accuracy curve $\text{Acc}@ \tau$ across detection thresholds $\tau \in [0.1, 0.9]$, contrasting reconciled X-first vs. unreconciled Y-first decoding; (b) Empirical IoU distribution histogram yielding 0.3671 mean IoU.

IX. CONCLUSION

We presented **SatQuery AI**, a parameter-efficient, edge-deployable vision-language architecture for unified Earth Observation intelligence. By combining zero-shot semantic intent routing with 4-bit NF4 quantized LoRA adaptation of Qwen2.5-VL-3B, SatQuery achieves high conversational reasoning (77.0% VQA accuracy) and spatial localization (0.3671 mean IoU, $\text{Acc}@0.5$ 44.0%) on held-out benchmarks within a ~ 2.4 GB memory buffer on an 8GB-class consumer GPU (architecturally an RTX 4060 laptop GPU; reported latencies captured via a verified remote GPU tunnel, Sec. VII). Reconciling coordinate serialization discrepancies yielded an immediate 17.1-fold improvement. SatQuery provides a practical blueprint for deploying multimodal remote sensing intelligence on consumer workstations.

ACKNOWLEDGMENT & OPEN SCIENCE

This research was developed under the Smart India Hackathon 2026 for the Indian Space Research Organisation (ISRO Problem Statement SIH26167). All model checkpoints, evaluation code, and benchmark logs are openly available at <https://github.com/sameelkazi/satquery>.

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